

The estimation ladder: configs 2, 3 and 4

*What each step of the modem chain adds to the phase-noise EVM, and what removes it
pn_cpe_study, 40 MHz 11ax/be, shipped LO, single LO, 8 frames, N_p = 6 pilots, 484 active tones*

Configs 2, 3 and 4 form a chain in which each step adds exactly one kind of error:

- 2 genie CPE, true channel — only the ICI the phase noise itself leaves.
- 3 2 + CFO acquisition (LTF pair, then the pilots' phase slope) + LTF channel estimate — plus one frozen estimation error.
- 4 3 with the genie CPE replaced by the N_p-pilot CPE — plus one estimation error that is independent per symbol.

Readings

	$\Delta f = 0$	$\Delta f = 2 \text{ kHz}$	vs the previous step ($\Delta f = 0$)
2	-44.20 dB	-26.74 dB	—
3	-42.47 dB	-42.47 dB	+1.73 dB
4	-42.14 dB	-42.14 dB	+0.33 dB

Three errors of different nature

Within one symbol the equalized subcarriers are $X_k(1 + \epsilon_k) + \text{ICI}_k$, where ϵ_k is the frozen error of the LTF channel estimate and ICI_k this symbol's phase-noise ICI. Config 2 reads σ^2_{ICI} alone. The channel estimate divides the LTF symbol's own ICI into $\hat{H}_k$, so every symbol of the packet pays it again:

$$\frac{P_{e,3}}{P_{e,2}} \approx 1 + \frac{\rho}{2} \quad (\text{two LTF repeats averaged, } \rho \approx 1) \Rightarrow +1.76 \text{ dB theory, } +1.73 \text{ dB measured}$$

An LS phase estimate over N subcarriers has error variance $(\sigma^2_{\text{ICI}} + \sigma^2_{\epsilon})/(2N)$; applied to all of the symbol's data points it adds that fraction of error power common-mode:

$$\frac{P_{e,4}}{P_{e,3}} \approx 1 + \frac{1}{2N_p} - \frac{1}{2N_{\text{active}}} \Rightarrow 10\log_{10}(1 + 1/12) = 0.35 \text{ dB theory, } +0.33 \text{ dB measured}$$

	2: ICI floor	3 – 2: frozen LTF error	4 – 3: pilot noise
averages with	nothing	the number of LTFs	the number of pilots
with symbols?	—	no (frozen per packet)	no (common-mode)
vs residual Δf	adds $1 - \text{sinc}^2(\Delta f T)$	unchanged	unchanged
vs LO level	follows it	constant dB gap	constant dB gap
vs symbol length T	grows with T	ratio unchanged	ratio unchanged

Three things worth noting

- Step 3 is the second-largest term, an order of magnitude above the pilot term. Intuition focuses on 'are there enough pilots', but the real cost sits in the channel estimate: the LTF is received once, its share of ICI is frozen into $\hat{H}$, and every symbol of the packet pays for it again. This is where 'phase noise's EVM contribution roughly doubles in the modem form' comes from ($1.5\times \rightarrow 1.76$ dB; the two LTF repeats' ICI is correlated at low offset, hence the slightly larger measured value).
- Config 2 is the only one that depends on Δf , because it has no acquisition stage; configs 3 and 4 read identically at $\Delta f = 0$ and 2 kHz. The tracking's value shows in the Δf -dependence of the $2 \rightarrow 3$ step; the estimation's cost shows in the two $\Delta f = 0$ steps of $2 \rightarrow 3 \rightarrow 4$ — one chain, two readings.
- Each term has its own remedy. Config 2 only through the LO (k_2 , loop bandwidth) or a shorter symbol. Step 3 through more LTFs (each de-rotated by its own CPE before averaging, otherwise the magnitude is biased), smoothing $\hat{H}$ across tones on a flat channel (here the channel is exactly unity, so tone-averaging would drive it to zero; a real frequency-selective channel limits it by its coherence bandwidth), or decision-directed channel tracking. Step 4 through N_p , pilot power (this model runs pilots at data power) or decision-directed CPE. The three do not add into one 'total gain', but each step of the staircase can be attributed on its own.

Expected pilot step by bandwidth

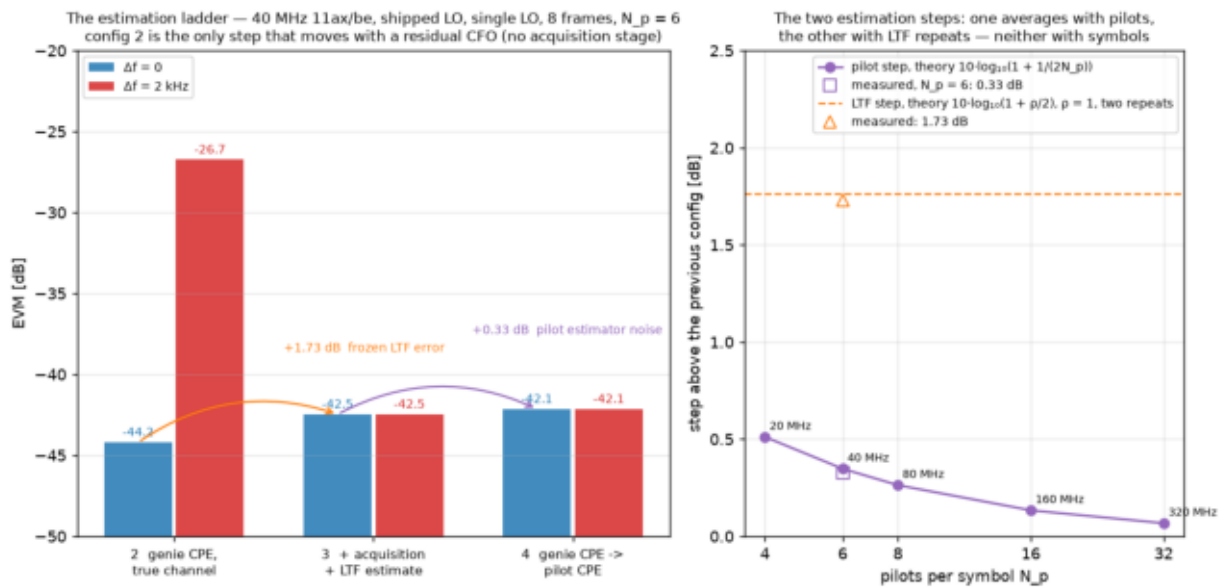
bandwidth	20 MHz	40 MHz	80 MHz	160 MHz	320 MHz
N_p	4	6	8	16	32
$10 \cdot \log_{10}(1 + 1/(2N_p))$	0.51 dB	0.35 dB	0.26 dB	0.13 dB	0.07 dB

On an OFDMA 26-tone RU with two pilots the same term is 0.97 dB — the only place where it becomes the dominant one.

Where the numbers come from

All readings are computed by this script through the study's own helpers (`_pn_nominal` at 40 MHz / 11ax / single LO / 8 frames, seed 0); the 8-frame values scatter about 0.3 dB rms seed to seed (frozen LTF error, one realization per frame). The chain is the one shipped in `pn_cpe_study 0.7.14`: LTF-pair coarse CFO, pilot-slope fine CFO, LTF channel estimate, then genie or pilot CPE.

Figure: the ladder at $\Delta f = 0$ and 2 kHz; the two steps vs N_p



Left: configs 2/3/4 with and without a 2 kHz residual CFO — only config 2 moves. Right: the pilot step against the number of pilots (theory $10 \cdot \log_{10}(1 + 1/(2N_p))$ with the 40 MHz measurement) and the LTF step (theory for two averaged repeats with the measurement).